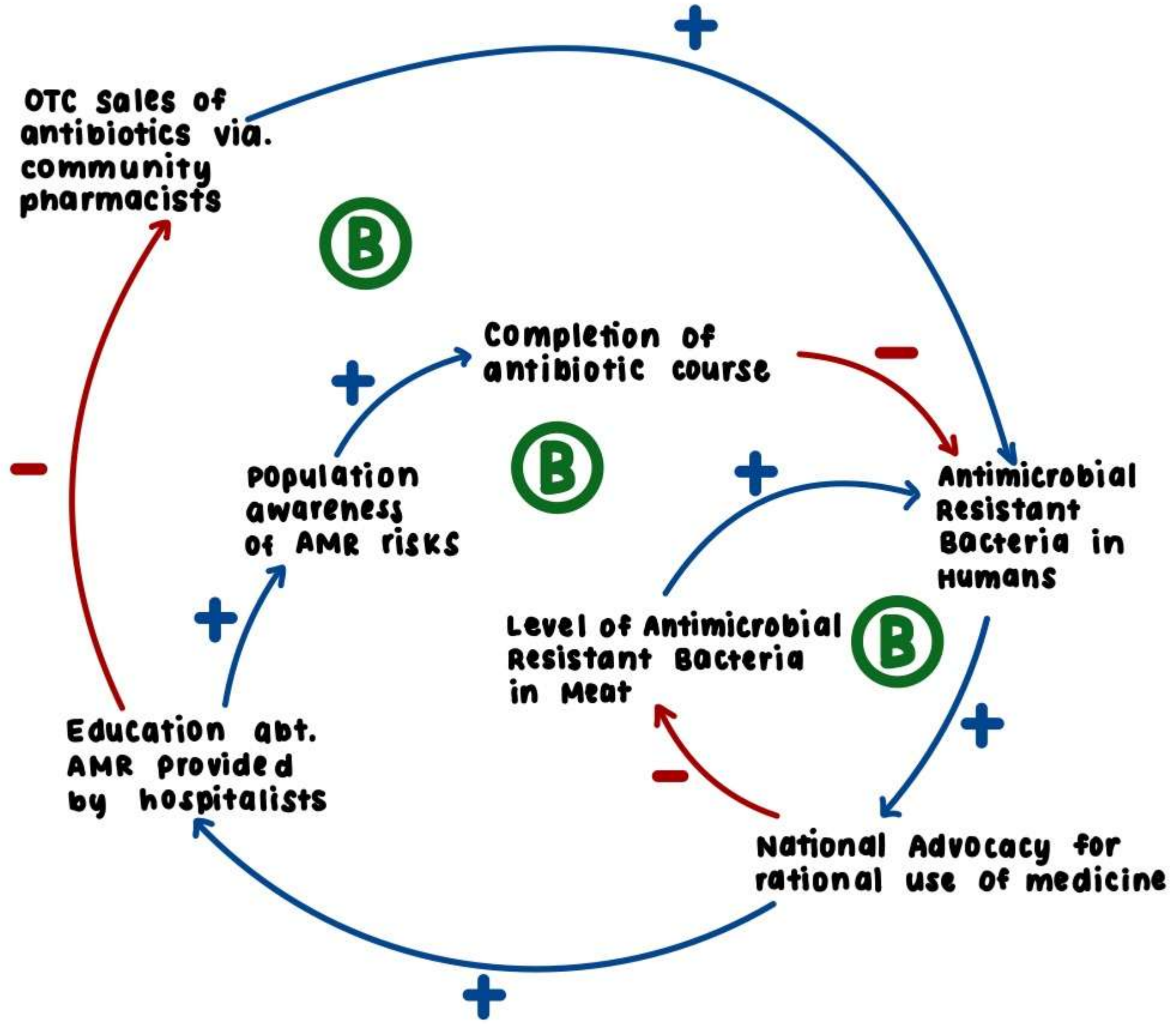
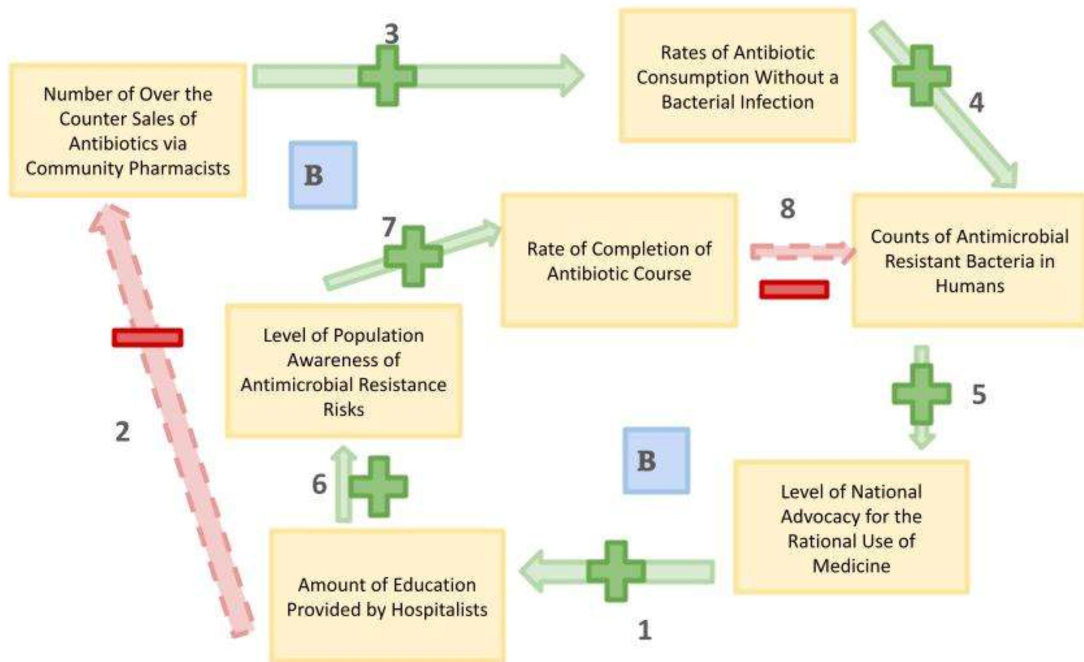
Part 1: Draft Map



Part 2: Revised Map



- Determinants Added:
    - Rates of Antibiotic Consumption Without a Bacterial Infection: I added this because it provided a clear connection between over the counter antibiotic sales and the counts of antimicrobial resistant bacteria in humans. This is because there are consistent observations in Thailand that community pharmacists do not sufficiently consider that infections may have viral causes, and dispense antibiotics to treat them[1], increasing antimicrobial resistance as a result, as the unnecessary use of antibiotics has been a leading cause for the rise in antibiotic resistance in the 90s and 2000s.[2] This shows how over the counter sales lead to consumption without a bacterial infection, leading to higher counts of antibiotic resistant bacteria.
- Determinants Removed:
    - Level of Antimicrobial Resistant Bacteria in Meat: I removed this because there was a lack of robustness in the causality between “Level of National Advocacy for Rational Use of Medicine” and this determinant. While there was an emphasis on meat and agriculture in the Thai national advocacy plans[3], there is insufficient evidence of these plans becoming legitimate legislation that impacted levels of antibiotic usage in meat.

Part 3: Relationship Narrative

1. As the level of national advocacy for the rational use of medicine increases through implementation of national plans, the amount of education provided to patients about antimicrobial resistance will increase. For example, one goal of the Thai RDU Hospital Project is to raise awareness and concerns about antimicrobial resistance among patients through hospital networking, and it successfully decreased mean antibiotic prescription rate from respiratory infections from 41.5% in 2014 to 21.0% in 2019[4]. As 36.1% of patients reported doctors and 24.8% reported health workers as their primary information sources about AMR in 2017[5], implementing plans like the RDU Service Plan will increase education by hospitalists to the public.

2. Although 52.4% of patients visit hospitals first when seeking antibiotics, 98.0% will search for antibiotics from a secondary source when they are not prescribed antibiotics at the first place they visit, and 52.6% of the time, the secondary source is a "clinic", which community pharmacies are under the umbrella of.[6] As 36.1% of patients reported doctors and 24.8% reported health workers as their primary information sources about AMR in 2017[5], increasing education by hospitalists increases knowledge about drug use. Patients with knowledge measured as "Adequate" had 3.37 times the odds of exhibiting rational drug use behavior compared with those measured as having "Inadequate" knowledge[6], which decreases the rate of seeking antibiotics from community pharmacies.

3. Over the counter and non-prescribed antibiotic use is associated with inappropriate drug and dose choices in patients.[7] For example, it is consistently observed in Thailand that community pharmacists do not sufficiently consider that infections may have viral causes, and dispense antibiotics to treat them, increasing antimicrobial resistance as a result.[1] This indicates that as over the counter sales of antibiotics increase, the rate of patients consuming antibiotics without a bacterial infection increases as well.

4. The unnecessary use of antibiotics has been a leading cause for the rise in antibiotic resistance in the 90s and 2000s.[2] Many medical associations, such as the American Dental Association, recommend professionals to "avoid unnecessary use of antibacterial drugs in treating viral infections" as a method of combating antibiotic resistance[8]. Therefore, as the rates of antibiotic use without a bacterial infection

increase, the counts of antibiotic resistant bacteria in people's bodies on average will increase in turn.

5. The burden of antimicrobial resistance, which includes up to 700,000 deaths globally per year[3], led to the global action plan on AMR being developed in 2015 by all the World Health Organization (WHO) member states at the World Health Assembly, calling on each member state to develop a national action plan within the next 2 years. This included Thailand, which developed a national strategic plan on antimicrobial resistance during the second half of 2015[3]. In this way, the increasing burden of antimicrobial resistance over time led to actionable plans at the national level, increasing the level of national advocacy for the rational use of medicine.

6. 36.1% of patients reported doctors and 24.8% reported health workers as their primary information sources about AMR in 2017[5], and medical associations such as the American Dental Association insist that it is incumbent on healthcare providers to use antibiotics in an appropriate manner.[8] By increasing the amount of education provided by hospital workers, encompassing doctors and other health workers such as nurses, it is possible to consequently increase the level of knowledge regarding the risks associated with antimicrobial resistance among the population, as these workers are primary source of information on the subject.

7. Patients with knowledge measured as "Adequate" had 3.37 times the odds of exhibiting rational drug use behavior compared with those measured as having "Inadequate" knowledge[6], and the WHO definition of rational drug use behavior includes the usage of medication for a sufficient duration of time[6]. Therefore, increasing a person's knowledge of AMR risks makes them more likely to consume their antibiotics for a sufficient duration of time.

8. In mathematical models, it is shown that early interruption of antibiotic therapy provides an advantage for antibiotic resistant strains of bacteria and results in infections caused by resistant bacteria to progress.[9] Medical associations like the American Dental Association recommend professionals to "stress the importance of completing the full course of therapy (that is, taking all doses for the prescribed treatment time)" as a method of reducing antibacterial resistant bacteria in patients.[8] Therefore, as rates of antibiotic course completion increase, we decrease

the early interruption of antibiotic course completion and in turn, decrease the counts of antibiotic resistant bacteria in people's bodies.

Part 4: Reflection

This assignment was a fascinating culmination of the content I learned in class and the research I did on the topic of antimicrobial resistance in Thailand. Prior to doing research on the subject, I thought the most prominent determinant would be the level of individual knowledge regarding the risks of antimicrobial resistance among the population, as I felt that it was due to deficiencies in cultural understanding of the topic that would lead to inappropriate usage of antibiotics. However, upon doing research, I discovered the critical role that community pharmacies play in the dissemination of antibiotics and information about their associated risks, as there is no legal restriction on the sale of antibiotics in Thailand like other countries, making informal purchase and consumption highly accessible.

The global health system structure I learned about in class was a massive influence on my map, as the national plans developed as part of my determinant "Level of National Advocacy for Rational Use of Medicine" were heavily influenced by the WHO identification of the severe health risks associated with antimicrobial resistance and proposals for solutions. In addition, I applied the upstream downstream principles I learned in class to further edit my map and make it as robust as possible.

The most significant revision I made to my draft map was to add the determinant "Completion of Antibiotics Course." I added this determinant, despite it initially only being in the "Connections to Other Determinants" column of my table, because it provided a clearer causal relationship between "Level of Population Awareness of AMR Risks" and "Counts of Antimicrobial Resistant Bacteria in Humans." Applying the upstream downstream principles, it was evident that while population awareness was upstream of antimicrobial resistant bacteria counts, there were numerous behaviors resulting from this awareness, most prominently the completion of antibiotics courses, that more directly impacted the amount of antibiotic resistant bacteria present in humans.

The greatest change I made from my draft map to my final map was removing the determinant "Level of Antimicrobial Resistant Bacteria in Meat." While consumption of meat treated with subtherapeutic doses of antibiotics is associated with development of antimicrobial resistant bacteria in the body, there is insufficient evidence of national advocacy for rational drug use impacting antibiotic use in Thai meat production. Narrowing the map to the remaining determinants led me to view antimicrobial resistance through the ways in which antibacterial drugs and the information surrounding them are handled and distributed.

After removing that determinant, I added “Rates of Antibiotic Consumption Without a Bacterial Infection” to my final map. Viewing antimicrobial resistance through this new lens, it was important to recognize the significant role that consuming antibiotics without a bacterial infection plays, as this behavior directly results from a combination of inadequate education and irrational consumption, both strongly influenced by the presence of community pharmacies in Thailand. In this way, I created a more robust causal relationship between “Over the Counter Sales of Antibiotics via Community Pharmacies” and “Counts of Antimicrobial Resistant Bacteria in Humans.”

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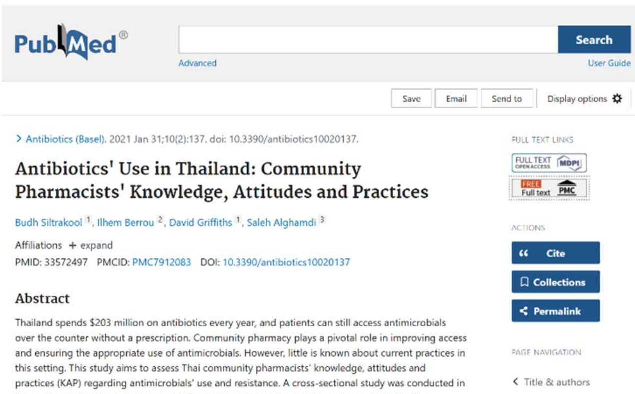
1.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FULL TEXT LINKS

Antibiotics' Use in Thailand: Community Pharmacists' Knowledge, Attitudes and Practices

Budh Siltrakool [1], Ilhem Berrou [2], David Griffiths [1], Saleh Alghamdi [3]

Affiliations + expand

PMID: 33572497 PMCID: PMC7912083 DOI: 10.3390/antibiotics10020137

ACTIONS

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Abstract

Thailand spends $203 million on antibiotics every year, and patients can still access antimicrobials over the counter without a prescription. Community pharmacy plays a pivotal role in improving access and ensuring the appropriate use of antimicrobials. However, little is known about current practices in this setting. This study aims to assess Thai community pharmacists' knowledge, attitudes and practices (KAP) regarding antimicrobials' use and resistance. A cross-sectional study was conducted in Bangkok and Chonburi province in 2017 using an online self-administered questionnaire. The

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Given that antimicrobial resistance is largely driven by the inappropriate use of antimicrobials, pharmacists are uniquely positioned to improve the utilisation of antimicrobials and preserve their efficacy. Community pharmacists in particular are key drivers to improving antimicrobial stewardship in contexts like Thailand, where pharmacies are the most cost-effective, accessible and sometimes the only healthcare provider available [10,11,12,13]. Evidence support the impact of pharmacists' interventions on reducing antimicrobial resistance [12,14,15,16]. However, evidence also indicate that in countries and settings where antimicrobial resistance rates are soaring, pharmacists may lack awareness and can have inadequate knowledge of antimicrobial resistance. For example, community pharmacists do not frequently consider that infections encountered in the community often have viral causes, and dispensing antibiotics to treat them will further increase antimicrobial resistance. This has been reported in Thailand [17,18] and other countries in the region [19].

2.


Nursing Clinics of North America

Volume 40, Issue 1, March 2005, Pages 63-75

Antibiotic Resistance

Maria A. Smith DSN, RN, CCRN

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FEEDBACK

Action plan for antibiotic resistance

The unnecessary use of systemic antibacterials has been a leading cause for the rise in antibiotic resistance in the past 5 to 10 years. This rise has occurred in both hospital and community-acquired infections. First steps to addressing the problem of antibiotic resistance are awareness and education of health care providers and the public. Education of health care providers about the hazards of prescribing antibiotics for illnesses that are nonbacterial will go a long way in decreasing the prevalence, and slowing the spread, of antibiotic resistance.

3.

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Public Health. 2018 Apr:157:142-146. doi: 10.1016/j.puhe.2018.01.005. Epub 2018 Mar 20.

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Abstract

Implementing national strategies on antimicrobial resistance in Thailand: potential challenges and solutions

A Sommanustweechai [1], V Tangcharoensathien [2], K Malathum [3], N Sumpradit [4], N Kiatying-Angsulee [5], N Janejai [6], S Jaroenpoj [7]

Affiliations + expand

PMID: 29524812 DOI: 10.1016/j.puhe.2018.01.005

Abstract

Background: Thailand has developed a national strategic plan on antimicrobial resistance (NSP-AMR) and endorsed by the Cabinet in August 2016. This study reviewed the main contents of the NSP-AMR and the mandates of relevant implementing agencies and identified challenges and recommends

Background

Antimicrobial resistance (AMR) is recognized as a major global human security threat. AMR affects human and animal health and economies worldwide with an estimated 700,000 AMR-attributable deaths annually.[1]

The global risk of AMR requires a comprehensive and integrated approach at global, regional, and country levels applying the 'One Health' principle. The global action plan on AMR was adopted in 2015 by all the World Health Organization (WHO) member states at the World Health Assembly, calling on each member state to develop a national action plan in 2 years.[2] The World Organization for Animal Health (OIE)[3] and the Food and Agricultural Organization[4] also adopted AMR resolutions in 2015, fostering tripartite collaboration.

Thailand developed a national strategic plan on AMR (NSP-AMR) during the second half of 2015, based on a strong participatory process. The NSP-AMR, a 5-year plan from 2017 to 2021, was endorsed by the Cabinet resolution in 2016.[5] Given the mandates assigned to implementing agencies, this article

4.

Pharmacy Practice
www.pharmacypractice.org

Pharm Pract (Granada). 2021 Feb 9;19(1):2201. doi: 10.18549/PharmPract.2021.1.2201

Effects of a national policy advocating rational drug use on decreases in outpatient antibiotic prescribing rates in Thailand

Onanong Waleekhachonloet [1], Thananan Rattanachotphanit [2], Chulaporn Limwattananon [3], Noppakun Thammatacharee [4], Supon Limwattananon [5]

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PMCID: PMC7886315 PMID: 33628347

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Abstract

INTRODUCTION

METHODS

RESULTS

Abstract

Objective:

This study examined the effects of a national policy advocating rational drug use (RDU), namely, the 'RDU Service Plan', starting in fiscal year 2017 and implemented by the Thai Ministry of Public Health (MOPH), on trends in antibiotic prescribing rates for outpatients.

	Antibiotics prescribing rate, mean[b] (standard deviation) [number of visits][c]
Respiratory infections[d]	
Fiscal year 2014	41.5 (14.6) [37, 344564]
Fiscal year 2015	40.5 (13.0) [78, 693546]
Fiscal year 2016	36.7 (12.2) [109, 1088750]
Fiscal year 2017	31.3 (11.0) [136, 1219255]
Fiscal year 2018	24.4 (8.4) [135, 1170717]
Fiscal year 2019	21.0 (6.7) [136, 1086206]

5.

Knowledge and use of antibiotics in Thailand: A 2017 national household survey

Sunicha Chanvatik, Hathairat Kosiyaporn, Angkana Lekagul, Wanwisa Kaewkhankhaeng, Vuthiphan Vongmongkol, Apichart Thunyahan, Viroj Tangcharoensathien

Published: August 9, 2019 • https://doi.org/10.1371/journal.pone.0220990

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Abstract
Introduction
Materials and methods
Results
Discussion
Conclusion
Supporting information
Acknowledgments
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Abstract

Background

The Thailand National Strategic Plan on Antimicrobial Resistance (AMR) 2017–2021, endorsed by the Thai Cabinet in 2016, aims to increase public knowledge about antibiotics and AMR awareness by 20% by 2021. This study assesses the prevalence of antibiotics use, clinical indications and sources; knowledge and access to information related to antibiotics and AMR; and factors related to level of knowledge and access to information among Thai adult population.

Methods

An AMR module was developed and embedded into the 2017 Health and Welfare Survey; a cross-sectional, two-stage stratified sampling, nationally representative household survey carried out biannually by National Statistical Office. The survey applied a structured interview questionnaire. The survey was conducted in March 2017 where 27,762 Thai adults were interviewed of the AMR module. Data were analyzed using descriptive and inferential statistics.

Results

Public information on appropriate use of antibiotics and AMR

Only 17.8% of respondents had received information about the appropriate use of antibiotics and AMR in last 12 months. Three common sources of information were doctors (36.1%), health workers (24.8%) and pharmacists (17.7%). Other sources such as conventional media (television and social media) played a minor role contributing 8.3% and 3.5% respectively, while 7.2% of respondents received information from family and friends (see Fig 4).

6.

Drug Healthc Patient Saf. 2022 Mar 25:14:27-36. doi: 10.2147/DHPS.S339808. eCollection 2022.

The Predictors Influencing the Rational Use of Antibiotics Among Public Sector: A Community-Based Survey in Thailand

Thaw Zin Lin [1], Isareethika Jayasvasti [2], Sariyamon Tiraphat [1], Supa Pengpid [1 3], Manisthawadee Jayasvasti [4], Phetlada Borriharn [5]

Affiliations + expand

PMID: 35369038 PMCID: PMC8965102 DOI: 10.2147/DHPS.S339808

Abstract

Background: The spread and emergence of antimicrobial resistance is the significant public health concerns over past decades. The major leading cause comes from irrational use of antibiotics.

Aim: To explore the characteristics of rational use of antibiotics and identify its predictive factors among public sector living in Nakhon Nayok Province, Thailand.

Methods: This project was conducted by using the data-source from Rational Use of Antibiotics (RUA) Survey Project at Nakhon Nayok Province. A cross-sectional community-based study method and face to face interviews were conducted. Two hundred fifty-four participants were selected by using Quota sampling method. Descriptive statistics were used to describe the sociodemographic and antibiotics use characteristics. Chi-square test were utilized to determine the association between

Antibiotic Access Sources/Antibiotic Usage Information Attainment Channels

Most respondents reported that their main sources of antibiotic (52.4%) attainment were hospitals and health promoting hospitals (HPH), followed by drugstore and grocery (35.4%). The majority (55.5%) firstly went to the hospitals and HPH when they felt sick and wanted to take antibiotics. Almost all of them (98.0%) replied that they would search for antibiotics when they were not prescribed antibiotics at the first place they visited and 56.2% chose 'clinics' as the second place to get antibiotics as depicted in Figure 1. Over 80% of the respondents searched for related information about antibiotic use from different channels, the majority (63.4%) searched for information from the physician or healthcare professionals at the hospitals.

7.

Meta-Analysis J Infect. 2019 Jan;78(1):8-18. doi: 10.1016/j.jinf.2018.07.001. Epub 2018 Jul 5.

Global access to antibiotics without prescription in community pharmacies: A systematic review and meta-analysis

Asa Auta [1], Muhammad Abdul Hadi [2], Enoche Oga [3], Emmanuel O Adewuyi [4], Samirah N Abdu-Aguye [5], Davies Adeloye [6], Barry Strickland-Hodge [7], Daniel J Morgan [8]

Affiliations + expand

PMID: 29981773 DOI: 10.1016/j.jinf.2018.07.001

Free article

Abstract

Objective: To estimate the proportion of over-the-counter antibiotic requests or consultations that resulted in non-prescription supply of antibiotics in community pharmacies globally.

Methods: We systematically searched EMBASE, Medline and CINAHL databases for studies published from January 2000 to September 2017 reporting the frequency of non-prescription sale and supply of antibiotics in community pharmacies across the world. Additional articles were identified by checking reference lists and a Google Scholar search. A random effects meta-analysis was conducted to calculate pooled estimates of non-prescription supply of antibiotics.

Strict implementation of restrictions on over-the-counter sales of antibiotics has been shown to be effective in reducing non-prescription antibiotic consumption in Brazil, Chile and South Korea and Mexico. [32, 33, 34] Given that many countries have laws prohibiting over-the-counter sales of antibiotics, there is a need to ensure that these laws are strictly enforced. However, it is important to understand that over-the-counter sale of some antibiotics may be supported in certain contexts. For example, in the UK, pharmacists legally supplied, using a patient group direction, azithromycin to patients with positive chlamydia test results. [35] Legislative changes in some countries including Canada and New Zealand have given designated pharmacists the authority to prescribe and dispense antibiotics for some conditions. For instance, in New Zealand, pharmacists can prescribe and dispense trimethoprim for short-term treatment of an uncomplicated UTI. [35]

8.

Guideline J Am Dent Assoc. 2004 Apr;135(4):484-7. doi: 10.14219/jada.archive.2004.0214.

Combating antibiotic resistance

American Dental Association Council on Scientific Affairs

PMID: 15127872 DOI: 10.14219/jada.archive.2004.0214

Erratum in

J Am Dent Assoc. 2004 Jun;135(6):727

Abstract

Background: The ADA Council on Scientific Affairs developed this report to provide dental professionals with current information on antibiotic resistance and related considerations about the clinical use of antibiotics that are unique to the practice of dentistry.

Overview: This report addresses the association between the overuse of antibiotics and the development of resistant bacteria. The Council also presents a set of clinical guidelines that urges dentists to consider using narrow-spectrum antibacterial drugs in simple infections to minimize disturbance of the normal microflora, and to preserve the use of broad-spectrum drugs for more complex infections.

Conclusions and practice implications: The Council recommends the prudent and appropriate use of antibacterial drugs to prolong their efficacy and promotes reserving their use for the management

4 avoid unnecessary use of antibacterial drugs in treating viral infections;

5 if treating empirically, revise treatment regimen based on patient progress or test results;

6 obtain thorough knowledge of the side effects and drug interactions of an antibacterial drug before prescribing it;

7 educate the patient regarding proper use of the drug and stress the importance of completing the full course of therapy (that is, taking all doses for the prescribed treatment time).

9.

The Impact of Different Antibiotic Regimens on the Emergence of Antimicrobial-Resistant Bacteria

Erika M C D'Agata [1,#], Myrielle Dupont-Rouzeyrol [2,#], Pierre Magal [3], Damien Olivier [4], Shigui Ruan [5,*]

Editor: Stefan Bereswill[6]

Author information • Article notes • Copyright and License information

PMCID: PMC2603320 PMID: 19112501

Abstract

Backgroud

The emergence and ongoing spread of antimicrobial-resistant bacteria is a major public health threat. Infections caused by antimicrobial-resistant bacteria are associated with substantially higher rates of morbidity and mortality compared to infections caused by antimicrobial-susceptible bacteria. The emergence and spread of these bacteria is complex and requires incorporating numerous interrelated factors which clinical studies cannot adequately address.

Antimicrobial exposure is central to the emergence and spread of antimicrobial-resistant bacteria. We have presented a model delineating the interrelated factors of antimicrobial therapy, the immune system and resistance gene transfer, and their effect on the emergence of antimicrobial-resistant bacterial strains. The model delineates novel findings with regards to the timing and sequence of antibiotic therapy in preventing the emergence of resistant strains. First, the model shows that shorter lengths of antibiotic therapy and early interruption of antibiotic therapy provide an advantage for the resistant strains and result in infections caused by these resistant bacteria to progress. Second, the model outlines the optimal antibiotic regimens which prevent the progression of infection with a resistant strain. Our model demonstrates that combination therapy with two antibiotics prevents treatment failure and the emergence of resistant strains, as opposed to sequential therapy. Third, the model shows that a delay in the start of therapy is one of the key factors in promoting the rapid rise in resistant strains. The early timing is even more important than the type of antibiotic regimen since early initiation of antimicrobials will prevent emergence of resistance regardless of whether antibiotics are administered sequentially or concurrently. Using a model system which incorporated population-level factors, D'Agata *et al.* [12] also demonstrated that early initiation of therapy is key to preventing the spread of antibiotic-resistant bacteria within a hospital setting.

AI Attestation

I attest that this submission did not use AI in its development or in the creation of any of its components.